

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0507

Robbie 138 kV
April 30, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0507. The Interconnection Customer proposes a 196.7 MW wind facility interconnecting at the Robbie 138 kV substation in the Pennsylvania area. The request was received on August 06, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,829,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Robbie - Kinder Morgan 138 kV	Bixby Hill - Tombstone 500 kV	195.2	152	51.3
Granada - Business Park 138 kV	Tombstone - Bixby Hill 500 kV	181.8	160	51.3
Granada - Kinder Morgan 138 kV	Tombstone - Bixby Hill 500 kV	155.7	192	51.3
Hamilton - Tombstone 138 kV	Ash St Sta 223 - Tombstone 345 kV	146.2	229	21.6
Bad Crk - Hamilton Substation 138 kV	Ash Street Sta 223 - Tombstone 345 kV Line	136.8	226	21.6
Bad Crk - Peterson 138 kV	Tombstone - Ash Street Sta 223 Substation 345 kV line	135.0	220	9.4
Peterson - Edwards St Substation 138 kV	Ash StSta 223 - Tombstone Substation 345 kV	133.5	200	9.4
Edwards St - Plainview Muncipal Power Substation 138 kV	Connell - Chiquita 345 kV line	110.4	190	5.2
Robbie Substation - Fourth Branch Hydroelectric Facility 138 kV	Granada - Kinder Morgan 138 kV	109.0	176	100.0
Cromby - Ash St Sta 223 345 kV Line	Moser Generating Station - Curecanti 345 kV	107.0	1,036	7.2
Halls - Bradford 138 kV	Ash St Sta 223 - Tombstone 345 kV	106.3	165	5.4
Harvey Couch - Mpc Generating Substation 138 kV	Hamilton - Tombstone 138 kV	106.1	181	6.5
Edwards Street - Bradford 138 kV	Tombstone - Ash St Station 223 345 kV L	100.1	199	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kinder Morgan - Robbie 138 kV	Rebuild line with higher-capacity conductor	\$8,427,000	100% (\$8,427,000)
Granada - Business Park 138 kV	Rebuild line with higher-capacity conductor	\$9,321,000	100% (\$9,321,000)
Granada - Kinder Morgan 138 kV	Rebuild line with higher-capacity conductor	\$1,982,000	100% (\$1,982,000)
Hamilton - Tombstone 138 kV	Rebuild line with higher-capacity conductor	\$5,024,000	86% (\$4,326,000)
Bad Crk - Hamilton Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,483,000	86% (\$1,277,000)
Bad Crk - Peterson 138 kV L	Rebuild line with higher-capacity conductor	\$1,302,000	65% (\$842,000)
Peterson - Edwards St 138 kV Line	Rebuild line with higher-capacity conductor	\$1,984,000	65% (\$1,284,000)
Edwards St - Plainview Muncipal Power Substation 138 kV	Rebuild line with higher-capacity conductor	\$4,706,000	34% (\$1,593,000)
Robbie Substation - Fourth Branch Hydroelectric Facility 138 kV	Reconductor line	\$4,400,000	100% (\$4,400,000)
Cromby - Ash St Sta 223 345 kV Line	Reconductor line	\$7,952,000	8% (\$617,000)
Halls - Bradford 138 kV	Reconductor line	\$1,112,000	59% (\$660,000)
Harvey Couch - Mpc Generating 138 kV	Reconductor line	\$2,855,000	100% (\$2,855,000)
Edwards Street - Bradford 138 kV line	Reconductor line	\$2,098,000	59% (\$1,245,000)
	Total allocated to project		\$38,829,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.